AquaTrace

Software Requirements Specification

Product and Engineering Specification

28 October 2026

EVIDENCE-FIRST INVESTIGATION SUPPORT | DOCUMENTATION ONLY

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1. Cover Page and Document Control

Document: AquaTrace Software Requirements Specification (SRS)
Version: 1.0-draft
Date: 28 October 2026
Status: Implementation specification draft; repository implementation not verified
Audience: Coding agent, software engineers, product owner, epidemiology reviewer, UI/UX implementer, test engineer, deployment owner

Control	Value
Product baseline	AquaTrace task brief supplied for this document
Repository inspected	No project repository or existing `docs/` directory was mounted in the available workspace
Existing implementation	Unverified: no source files, lockfiles, environment files, datasets, or UI references were available
Requirement status	`Proposed` = specified but not proven implemented; `TBD` = approval/evidence required; `Verified` = observed reproducible check only
Scope	Documentation only; this deliverable does not implement or modify the application
Document owner / reviewers	TBD

> **Safety-critical rule:** AquaTrace supports investigation prioritisation. Statistical association is not confirmation that a source is contaminated. No screen, ranking, AI summary, or report may state otherwise without independent, documented confirmation evidence reviewed by an authorised officer.

2. Executive Summary

AquaTrace is an evidence-first investigation workstation for authorised health officers and field epidemiology teams. It imports participant-level illness and water-source exposure data, validates and versions those data, computes reproducible source-by-source comparisons, supports historical replay based on when evidence became available, and exports an investigation brief tied to a specific analysis snapshot. Optional AI text generation may explain already-computed findings or organise reviewed field notes; it cannot perform authoritative calculations or make operational decisions.

The established technology direction is Next.js, FastAPI/Python, MongoDB, and optional LangChain/LangGraph orchestration with a configured Groq API model. Exact framework/library versions, authentication provider, hosting, map provider, PDF renderer, and configured model were not available for repository inspection; those implementation facts remain unverified.

The MVP is complete when a user can create an investigation, validate/import a CSV into an immutable dataset version, inspect data quality, obtain deterministic per-source 2-by-2 analysis, replay only records available at a selected historical cutoff, and produce a matching PDF/print-ready field brief. Maps and Response Planner are conditional extensions and must not block that workflow.

2.1 Normative language

- **MUST**: required for MVP or safety/integrity.
- **SHOULD**: expected unless a reviewed reason prevents it.
- **MAY**: optional, explicitly scoped capability.
- **Proposed default**: an implementation recommendation, not an already approved/deployed choice.

- **TBD**: a decision requiring evidence or owner approval.

3. Purpose and Scope

3.1 Purpose

This SRS is the implementer-facing contract for AquaTrace. It defines user-visible behaviour, API responsibilities, data contracts, deterministic calculations, security boundaries, design rules, performance targets, tests, and completion criteria. A coding agent MUST follow it or raise a documented conflict before changing the contract.

3.2 In scope

1. Investigation creation, selection, metadata, provenance, and dataset-version history.
2. CSV validation, preview, fatal/blocking errors, warnings, and downloadable validation reports.
3. Per-source 2-by-2 tables, attack rates, relative risk (RR), 95% confidence intervals when supported, limitations, and transparent candidate-priority labels.
4. Reproducible historical evidence replay using `recorded_at <= as_of`.
5. Data-quality reporting, source metadata, optional validated source map, and field brief export.
6. Bounded optional LLM-generated explanations using aggregate/de-identified context only.
7. Investigation access isolation, privacy, safe errors, structured logs, and appropriate auditability.
8. Optional Response Planner after the core analysis-to-report path is stable.

3.3 Out of scope

- Diagnosis, treatment advice, individual clinical decision support, laboratory interpretation, or declaring contamination without independent confirmation.
- Person-level causal inference, confounder-adjusted analysis, geospatial exposure assignment, nearest-source attribution, exclusive source assignment, or automated outbreak declarations.
- Automatic emergency dispatch, water distribution, supply ordering, or opaque resource optimisation.
- Model training/fine-tuning, GPU training infrastructure, or claims of a custom trained model.
- Public release of identifiable participant-level data or claims that a prototype is certified/production-ready.

4. Product Vision and Problem Statement

Investigators may face several possible water sources, incomplete and changing exposure information, limited sampling resources, and pressure to decide what to investigate first. Isolated case counts and unvalidated spreadsheets do not provide a consistent way to validate records, compare exposed/unexposed groups, account for uncertainty, track how evidence evolved, and document next steps. AquaTrace turns supplied evidence into a traceable investigation workflow while preserving unknown values, import history, methodological limitations, and human decision authority.

AquaTrace is an operational investigation tool, not a generic AI dashboard or chatbot wrapper. The default experience foregrounds investigation context, provenance, participant counts, source comparisons, caveats, timeline, and next actions—not a prominent conversational interface.

5. Objectives and Measurable Success Criteria

ID	Objective	Acceptance measure	Verification evidence
OBJ-01	Reproducible analysis	Same immutable dataset version, cutoff and method version yield identical counts/ statistics	Unit/integration tests
OBJ-02	Safe unknown handling	Unknown illness/exposure never enters known 2-by-2 cells; counts remain visible	Data-quality/statistics tests
OBJ-03	Explainable prioritisation	Every label follows deterministic rules and contains rationale codes	Rule tests/source detail
OBJ-04	Historical integrity	No record with `recorded_at > as_of` influences any replay result	Timeline leakage tests
OBJ-05	Provenance	Each analysis/report names dataset version, provenance, cutoff and method version	API/report tests
OBJ-06	Degraded-mode usability	Groq failure does not block import, deterministic analysis, replay, or report	Failure-mode E2E
OBJ-07	Consistent interface	Four-token palette and central design tokens used across routes	CSS/contrast/visual checks
OBJ-08	End-to-end usefulness	Ordinary valid data can move from upload to exported brief without external spreadsheet editing	E2E test

These are proposed targets, not claims that any application code has met them. Performance acceptance targets appear in Section 20.

6. Definitions, Acronyms, and Terminology

Term	Definition
Attack rate	Proportion ill in a defined exposed or unexposed group during the specified period
Candidate source	Reported water source examined as a possible exposure, not a confirmed cause
CI	Confidence interval; here, nominal 95% interval for RR when method conditions allow
Dataset version	Immutable imported participant/source-exposure snapshot with validation summary and fingerprint
`recorded_at`	Timestamp when the record became available to the investigation; not survey date or symptom onset
Exposure status	`exposed`, `unexposed`, or `unknown` for each source independently
Illness status	Supplied classification `ill`, `not_ill`, or `unknown`
RR	Relative risk/risk ratio: exposed attack rate divided by unexposed attack rate
Synthetic data	Generated demo data, clearly labelled and not evidence about real people/places

Term	Definition
Aggregate reference	Published group-level counts; individual participant records do not exist in the reference table
Snapshot	Analysis result bound to an immutable dataset version and method/parameters
TBD	Not established by available evidence; needs a decision or verification

7. Stakeholders and User Classes

User class	Goals	Access expectation
Health officer / investigator	Inspect evidence, compare sources, document limitations, export brief, record follow-up	Read/update assigned investigations; create analyses/reports
Field epidemiology team member	Review import quality and source-level tables; supply follow-up observations	Assigned investigation access only
Response coordinator	Review brief/task status and approve next steps	Read and, if enabled, manage assigned tasks/resources
Administrator/deployment operator	Configure service, secrets, retention and access controls	Infrastructure/config access; not automatically participant-data access
Developer/test operator	Validate behaviour with synthetic fixtures	Synthetic data by default; real data only if explicitly authorised

The MVP MUST enforce investigation isolation. Complex enterprise roles are not required unless deployment context justifies them. Authentication provider and fine-grained role model are TBD; recommended default is OIDC-compatible identity with server-verified tokens and deny-by-default access. A public unrestricted demo mode MUST NOT be used with real health data.

8. Assumptions, Dependencies, and Constraints

8.1 Established product constraints

- Frontend: Next.js; backend API: FastAPI/Python; database: MongoDB.
- Optional LLM orchestration: LangChain/LangGraph; provider: Groq API using server-side key and explicit model configuration.
- Statistics are deterministic Python functions with automated tests; an LLM never calculates authoritative figures.
- Palette is limited to `#111827`, `#374151`, `#F3F4F6`, `#FFFFFF`.
- Two primary typography roles (Heading and Body), global stylesheet, responsive layout and lazy loading of appropriate heavy modules are mandatory.
- Synthetic demo data and published aggregate references must be separated.
- The provided 240-person marginals and CDC/Mehar references are targets, not proof fixture files exist.

8.2 Unverified environment facts

No AquaTrace repository, README, dependency manifest/lockfile, `.env.example`, dataset fixture, UI reference or deployment configuration was available in the mounted workspace. Exact package versions, actual route names, existing CSS, collection names, fixture status and implementation status cannot be claimed. Check package APIs against installed versions and official documentation when the repository is available.

8.3 Working assumptions

- One participant per CSV row; a unique pseudonymous participant ID; one exposure field per source.
- A participant may report multiple source exposures; analyses are source-by-source.
- Participant-level files are private health-related investigation data.
- `recorded_at` must be supplied/confirmed and parsed as a timezone-aware instant, then normalized to UTC.
- A validation session can stage the import privately before explicit confirmation. Proposed default validation-session TTL is 30 minutes; expiry invalidates commit and abandons the staged preview safely. Confirm against storage and user workflow during implementation.
- Performance values are targets to measure, not measurements already achieved.

9. Product Perspective and System Context

AquaTrace is a private web application with browser UI, FastAPI API, MongoDB persistence, optional external Groq service, and optional mapping provider. The report renderer creates a document from a validated analysis snapshot. External integrations are fallible; untrusted input is validated at server boundaries.

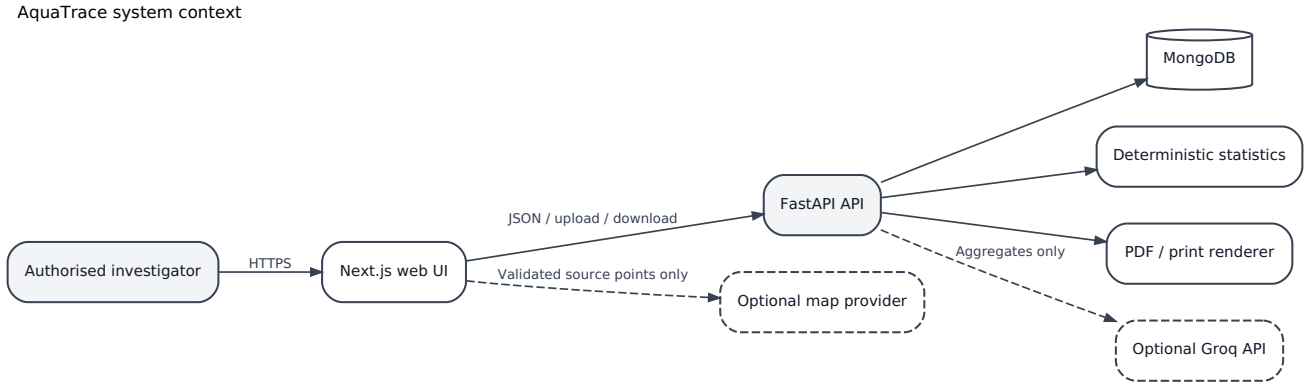


Figure 1. System context.

Map and LLM are optional; their failure cannot block deterministic analysis. Raw participant rows MUST NOT be sent to Groq.

10. Overall Product Architecture

10.1 Logical components

1. **Next.js UI:** pages, accessible forms, upload/preview, data-quality views, result presentation, timeline selection, optional lazy modules, downloads. It does not calculate authoritative statistics.
2. **FastAPI request layer:** authentication, authorization, request validation, safe errors, request IDs, rate limits and API routing.
3. **Import/validation:** file bounds, CSV decoding, schema checks, row issues, staging, explicit transformations, hashing and immutable version creation.
4. **Data-access layer:** scoped queries, indexes, persistence, version references, lifecycle and deletion policy.
5. **Deterministic engine:** source-specific 2-by-2 counts, rates, RR, CI eligibility, limitations, priority class and reproducibility key.
6. **Timeline service:** select records with `recorded_at <= as_of` before computing derived output.

7. **Report service:** render only the selected authorised snapshot; include provenance/limitations and private download handling.
8. **Optional AI service:** aggregate/de-identified allowlist context, one configured Groq model, output validation and deterministic fallback.
9. **Optional Response Planner:** manual tasks/inventory/assignments; no autonomous optimisation or dispatch.

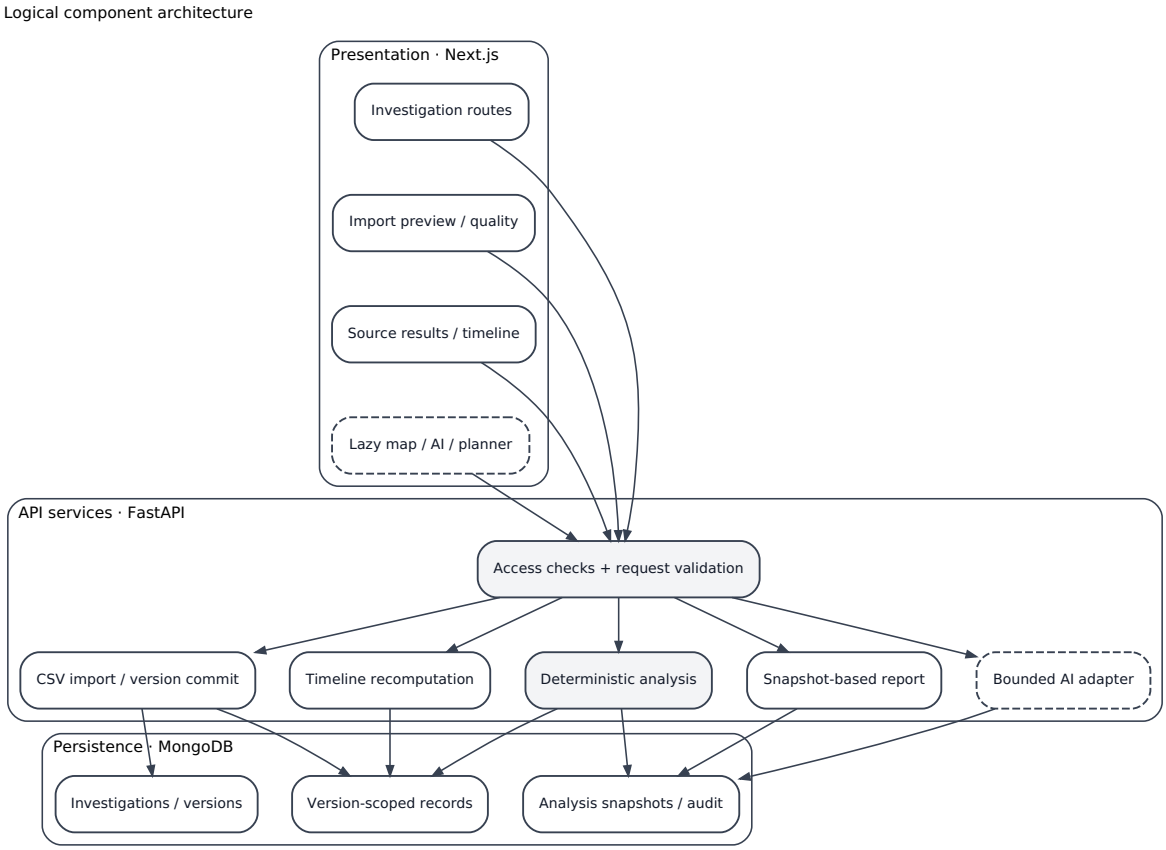


Figure 2. Logical component architecture.

Source calculations/report values belong to deterministic backend code; data access enforces investigation scope.

10.2 Trust boundaries

- Browser-to-API: all client values are untrusted; validate and authorize every resource ID server-side.
- API-to-MongoDB: scoped queries, driver parameters, schema validation and least-privilege credentials.
- API-to-Groq: only approved aggregates and de-identified notes; never participant IDs, raw CSV, direct identifiers or household coordinates.
- Browser-to-map: only authorised, supplied source coordinates; never participant/household points.
- Report download: re-check access to the underlying investigation and analysis before returning bytes.

10.3 CSV-to-report data flow

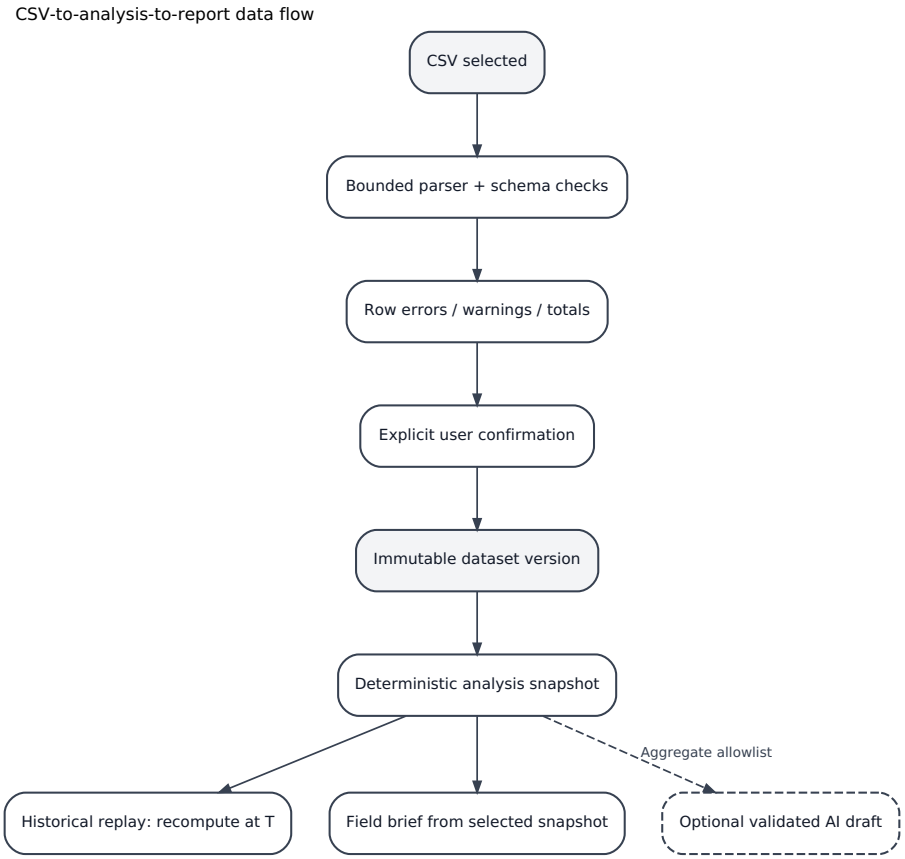


Figure 3. CSV-to-analysis-to-report data flow.

UI and PDF must use the same analysis snapshot and calculation-method version.

10.4 Document relationships

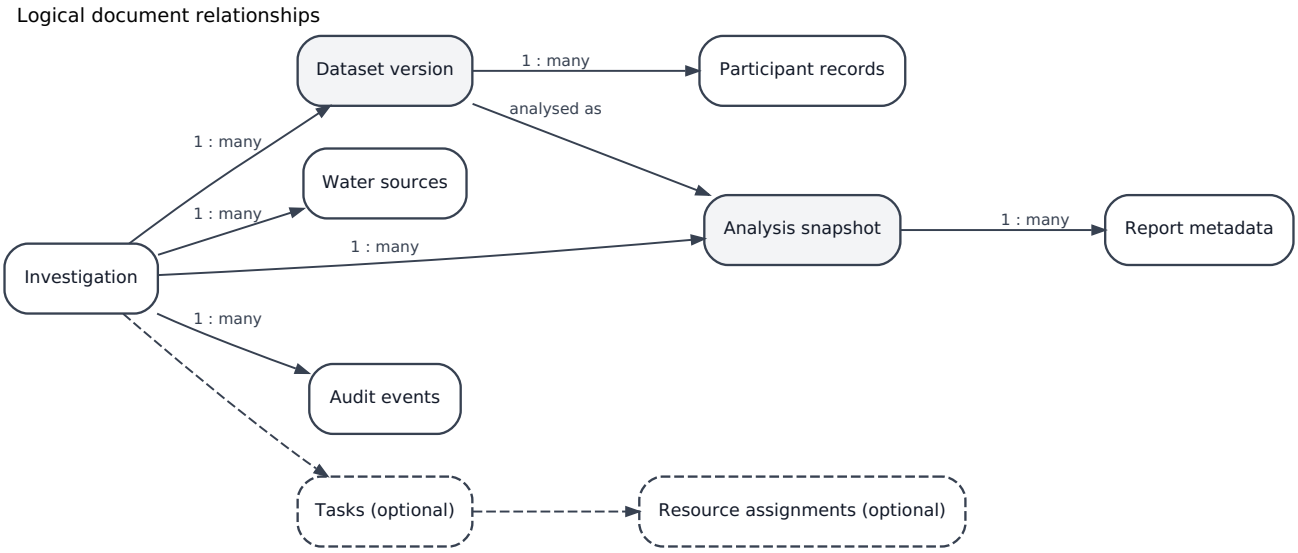


Figure 4. Logical document relationships.

This is a conceptual relationship diagram, not a claim that these collection names currently exist. See the Data Dictionary for proposed fields/indexes. Participant records are not embedded as an unbounded array inside an investigation document.

10.5 Deployment view

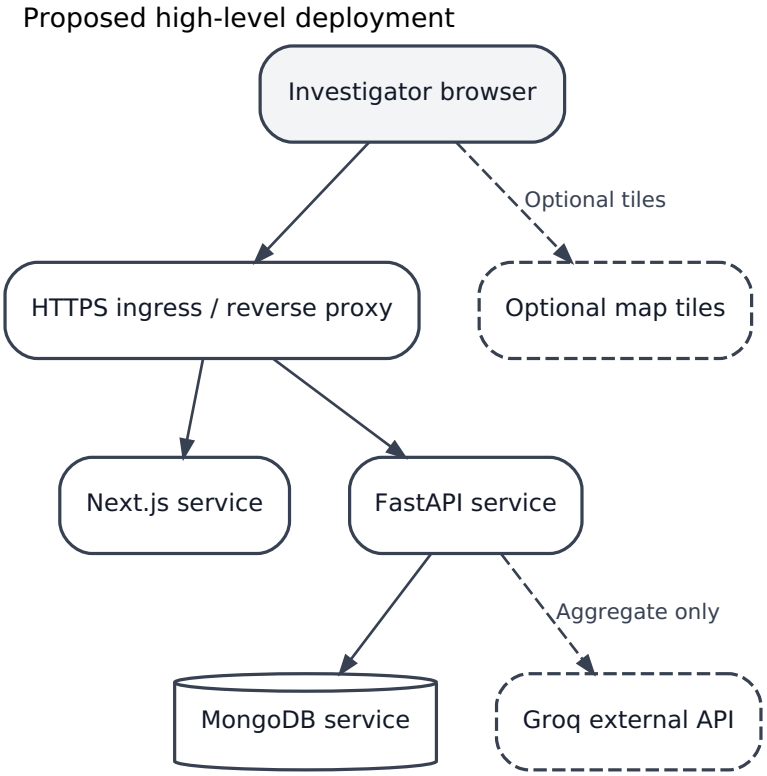


Figure 5. Proposed high-level deployment view.

Hosting, DNS, ingress, session configuration, database topology, and secret manager are TBD. MongoDB is not exposed to the browser. Groq and map integrations are optional external trust boundaries.

10.6 Optional Response Planner workflow

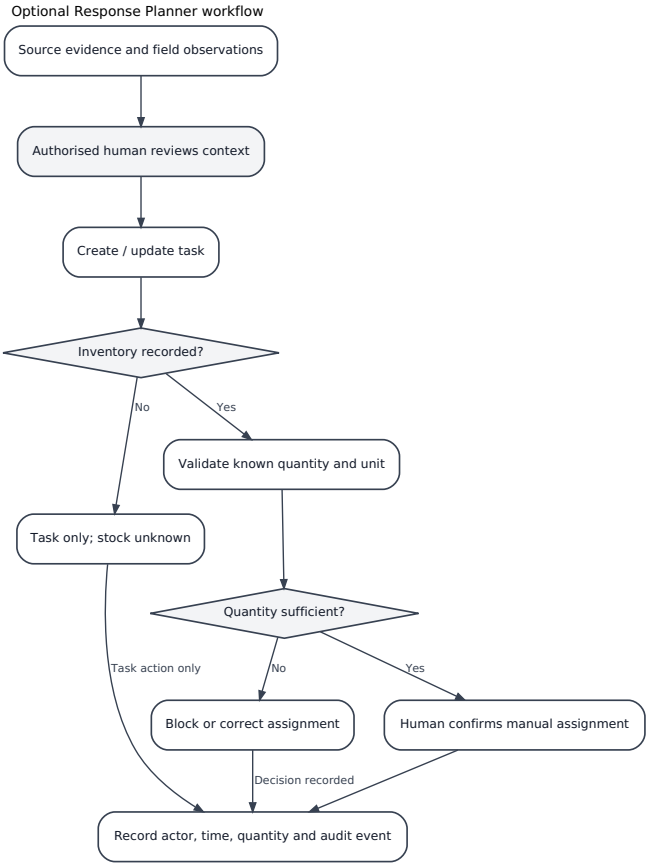


Figure 6. Optional Response Planner workflow.

Candidate source status alone must not trigger inventory assignment, water distribution, emergency orders or dispatch. When stock is unknown, the UI must not claim availability.

11. User Workflows and Use Cases

11.1 End-to-end workflow and state contracts

Step	UI action / component	Backend responsibility and data	States and completion evidence
1. Open workspace	Investigation list	Return authorised summaries only	Loading, empty-with-create-action, retryable error, ready list; no invented metrics
2. Create/select	New investigation form/list	Validate metadata and actor scope	Field errors; success opens investigation; deleted/ unavailable item returns safe not-found state
3. Review context	Investigation overview	Return provenance, current dataset version and history	Empty state explains no import exists; no implicit synthetic dataset
4. Select CSV	Import wizard	Enforce byte/type/encoding limits	File name/size, cancel/retry, malformed/oversized rejection
5. Validate/preview	Header mapping, sample rows, issue table	Parse into bounded private validation session	Progress; fatal/blocking/ warning severity; confirmation blocked by blocking errors

Step	UI action / component	Backend responsibility and data	States and completion evidence
6. Resolve issues	Validation issue list	No silent cleaning or row loss	All rows sharing a duplicate participant ID are rejected until corrected/re-uploaded or explicitly excluded; warnings acknowledged explicitly
7. Confirm import	Confirmation summary	Commit exactly one immutable version	Idempotent retry; partial failure never appears as success; version ID/hash/ counts shown
8. Inspect quality	Data-quality panel	Reconcile counts from committed version	Per-source unknown exposure counts and import warning counts visible
9. Run analysis	Run analysis action	Calculate for explicit dataset version/cutoff	Progress, safe failure; success identifies analysis ID/method
10. Compare sources	Table/chart	Render saved backend snapshot	Counts, rates, RR/CI/status/ caveats agree; unknown exposure separate
11. Source detail	Detail drawer/page	Return 2x2 table, rates, CI reason, flags and rationale	No Infinity/NaN; optional tasks/ coordinates only if present
12. Timeline replay	Timestamp control	Filter `recorded_at <= T`, recompute all derived values	Timezone/cutoff visible; no-records empty state; ranking changes attributable to eligible new evidence
13. Map	Lazy map only if source coordinates exist	Validate coordinates/ provenance	No coordinate fallback; provider retry/table fallback; synthetic points labelled; no household markers
14. Optional tasks	Task/resource panel	Persist manual records and audit changes	Known inventory conflicts block or require authorised reason; no dispatch
15. Export brief	Report request/download	Render a specified snapshot	Loading/error; private download; report version/cutoff and values match UI
16. Follow-up/update	Notes/task/new upload	Record provenance/audit and create new version for new data	Old snapshots remain intact; new version requires new analysis

11.2 CSV import sequence

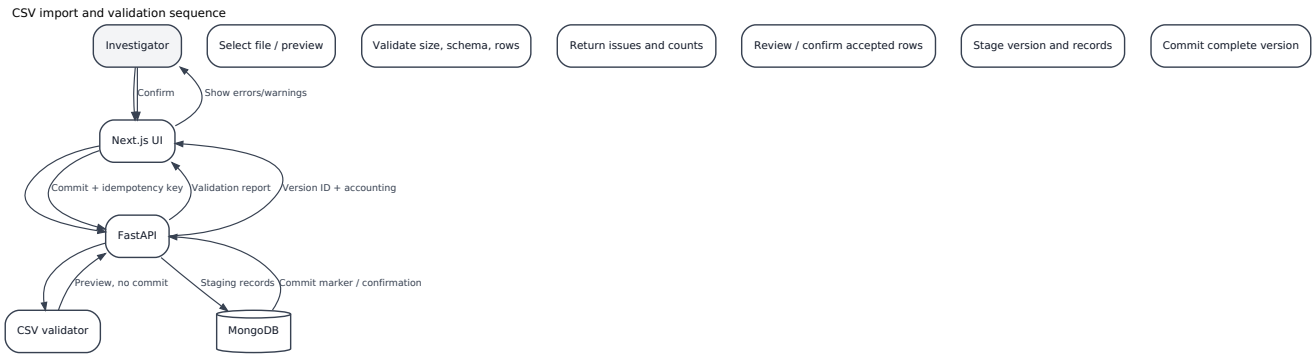


Figure 7. CSV import and validation sequence.

If persistence fails, the API must not report a committed version. Staging expiry and cleanup are mandatory.

11.3 Deterministic analysis sequence

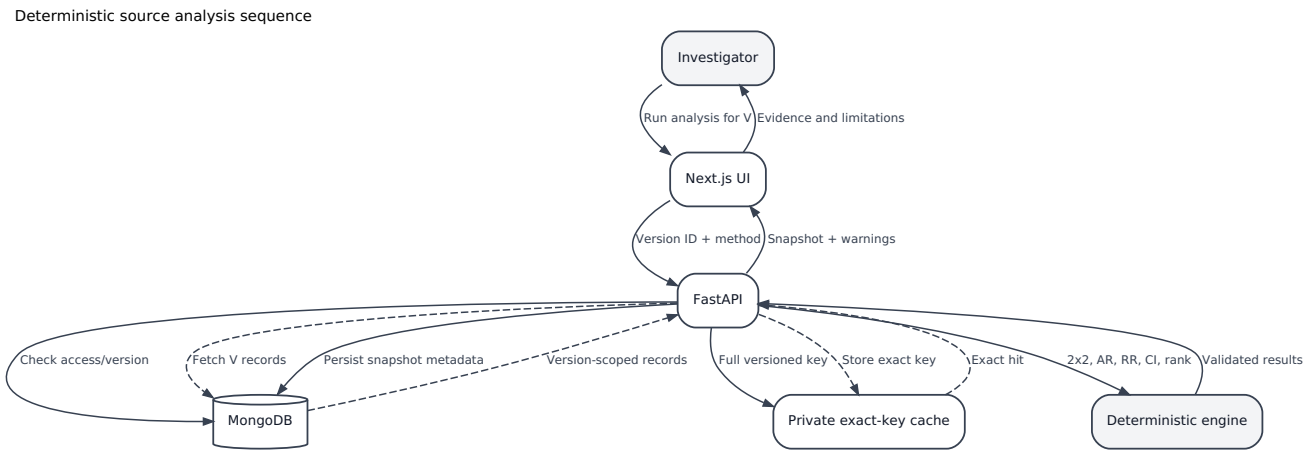


Figure 8. Deterministic source analysis sequence.

The frontend and LLM are not calculation authorities.

11.4 Timeline replay sequence

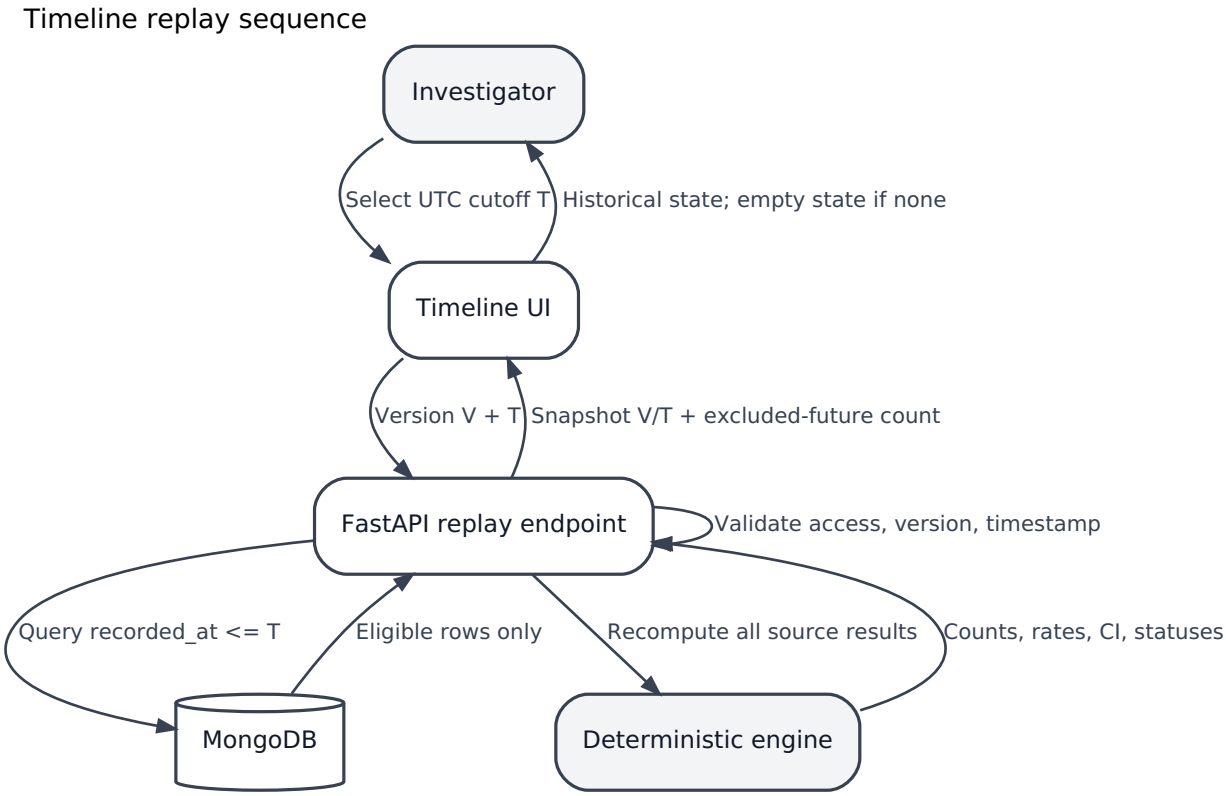


Figure 9. Historical timeline replay sequence.

Filtering only a displayed chart is insufficient; filter before every statistic/status is derived.

11.5 Optional AI summary sequence

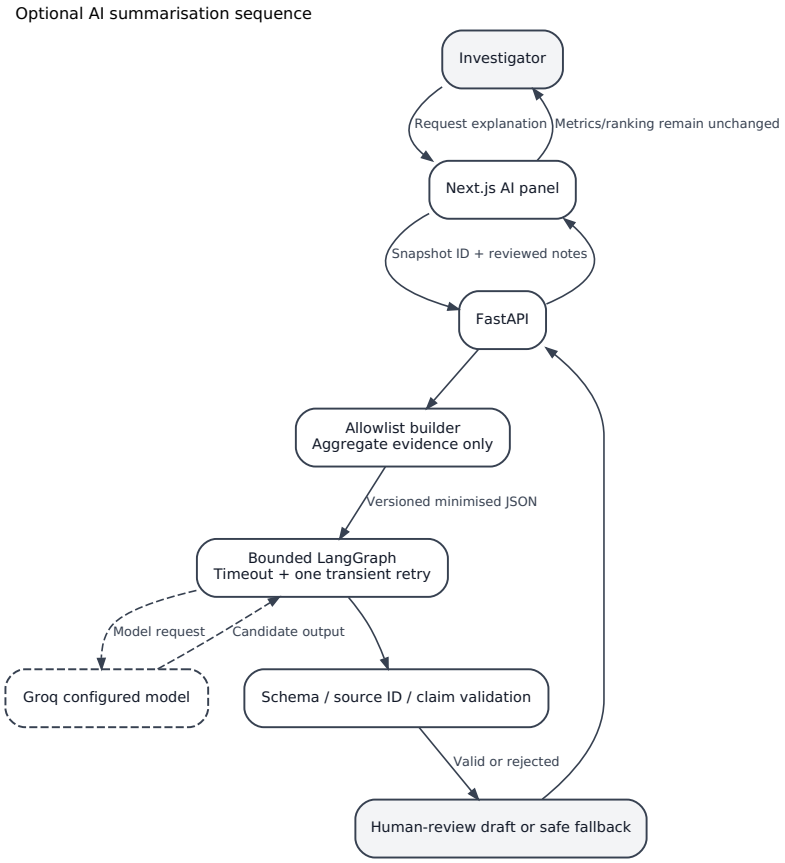


Figure 10. Optional AI summarisation sequence.

AI failure never invalidates or blocks a saved analysis snapshot.

12. Functional Requirements

The requirements below are proposed and unverified until application tests demonstrate them. Detailed case IDs are in `AquaTrace_Testing_and_Acceptance.md`.

12.1 Investigation and provenance

ID	Priority	Required behaviour	Verification
FR-001	MUST	Authorized user creates/lists/opens/updates an investigation with title, coarse area label, notes, status and provenance	API/UI/E2E
FR-002	MUST	Every child-resource request is scoped to server-verified access; unauthorized IDs reveal no participant data	API security
FR-003	MUST	Committed dataset versions are immutable; corrections create a new version	Integration
FR-004	MUST	Dashboard metrics derive from the displayed version/snapshot; absent values are not fabricated	UI/E2E

ID	Priority	Required behaviour	Verification
FR-005	MUST	Import/analysis history and provenance classes are visible; aggregate references are not person-level data	API/UI
FR-006	SHOULD	Archive/delete requires confirmation and an approved child-record retention policy	Integration/security

12.2 Import and data quality

ID	Priority	Required behaviour	Verification
FR-010	MUST	Server enforces file/row/column/source limits and safe parser time/space bounds	API/security/perf
FR-011	MUST	Server validates and returns row/file findings before commit	API integration
FR-012	MUST	Missing required headers and any unrecognised non-optional/non-exposure header are blocking schema errors; only documented optional fields and `exposure__<source_key>` columns are accepted	Unit/API
FR-013	MUST	Illness/exposure categories use explicit enums; invalid non-empty values block affected rows	Unit/API
FR-014	MUST	Unknown/blank exposure is never treated as unexposed; per-source unknown counts are shown	Unit/E2E
FR-015	MUST	Unknown illness is excluded from illness cells and counted separately	Unit/UI
FR-016	MUST	Duplicate participant IDs block confirmation until explicitly corrected/excluded; no silent deduplication	Import tests
FR-017	MUST	Invalid required timestamp rejects row; missing optional date warns; ambiguous dates are not guessed	Unit/API
FR-018	MUST	Exclusions, normalisations and acknowledged warnings are recorded; no silent data alteration	Integration
FR-019	MUST	Validation/rejected-row reports can be downloaded with safe CSV escaping	API/security
FR-020	MUST	Import confirmation is idempotent and commits one immutable version or fails visibly	Integration/failure injection
FR-021	MUST	Version stores fingerprint, schema/provenance, counts, source mapping and upload/commit time	Data/API

12.3 Analysis and interpretation

ID	Priority	Required behaviour	Verification
FR-030	MUST	Every source analysed independently; overlapping participant exposures remain possible	Unit test
FR-031	MUST	Source comparison displays `a,b,c,d`, denominators, rates, RR/CI where valid, unknown exposure and warnings	UI/API
FR-032	MUST	Undefined values are null plus reason; NaN/Infinity never reach JSON/UI/report	Unit/API
FR-033	MUST	Evidence classes/ranking use documented deterministic rules and reasons; LLM cannot change them	Unit/integration
FR-034	MUST	UI/report warns association is not confirmation and exposes confounding/missingness/ overlap limitations	Manual/UI/report
FR-035	MUST	Snapshot identifies dataset version/hash, cutoff, filter/ method version, results, warnings and timestamp	API/data
FR-036	SHOULD	Compare snapshots without mixing versions/cutoffs	Integration/UI

12.4 Timeline, map and report

ID	Priority	Required behaviour	Verification
FR-040	MUST	Replay includes only UTC `recorded_at <= as_of`, then recomputes all derived outputs	Regression tests
FR-041	MUST	Distinguish record availability, survey date, symptom onset and unknown contamination time	UI/report
FR-042	MUST	Show only explicit validated source coordinates with provenance; never infer coordinates/nearest-source exposure	Map tests
FR-043	SHOULD	Map is lazy-loaded; missing coordinates/provider failure leave source-table fallback	UI/perf
FR-044	MUST	Report uses an explicit authorized snapshot and matches screen counts/rates/ RR/CI/status/cutoff	Report integration
FR-045	MUST	Brief includes provenance, version, counts, unknowns, method, limitations, verification steps and disclaimer	Report acceptance
FR-046	MUST	Report generation/download uses access checks, safe names and private storage/ cache policies	Security/API

12.5 AI and optional planner

ID	Priority	Required behaviour	Verification
FR-050	MUST	Core workflow works without Groq key/service	Failure-mode E2E
FR-051	MUST	AI gets only allowlisted aggregate/de-identified data and reviewed notes, not raw rows/direct IDs	Payload/security review
FR-052	MUST	AI output passes versioned schema/source ID/length/ limitation validation	Unit/integration
FR-053	MUST	AI output is a human-review draft and cannot alter metrics/ ranking/assignments	UI/integration
FR-054	MUST	Provider failures/malformed output result in safe fallback and no secret/personal-data logs	Fault injection
FR-060	MAY, after core	Manual tasks have state, due date, responsible role, notes/ history	API/UI
FR-061	MAY, after core	Inventory/assignment enforces nonnegative quantities and warns/blocks exceeding known stock	Unit/API
FR-062	MUST if planner enabled	Manual assignment changes record actor/time; no automated dispatch/order	Audit/E2E
FR-070	MUST	Async pages expose loading/ empty/success/error states; critical content is eager	UI/E2E
FR-071	MUST	Stable safe API error envelope includes code, message, request ID and optional field errors	API/security
FR-072	MUST	Interrupted/repeated requests cannot create duplicate imports or claim uncommitted success	Integration

13. Frontend Requirements

13.1 Page inventory

Suggested routes (not observed routes): `/investigations`, `/investigations/new`, `/investigations/[id]`, `/investigations/[id]/import`, `/investigations/[id]/sources/[sourceId]`, `/investigations/[id]/timeline`, `/investigations/[id]/reports/[reportId]`, and optional `/investigations/[id]/tasks`. Follow the actual router pattern when available; do not replace an existing routing convention without need.

13.2 Dashboard content

Show investigation title, geographic label/status/date range if supplied; dataset version and provenance; row totals/import time; known ill/not-ill and unknown illness; per-source unknown exposure; source count and evidence-class distribution; recent quality warnings; last successful analysis; actions for import, analysis, comparison, timeline and brief export. Each number must be traceable to the version/snapshot named on screen. Do not mix current metadata with historical metrics without explicit labels.

13.3 Reusable components and interaction states

Recommended components: `InvestigationHeader`, `ProvenanceLabel`, `DatasetVersionSelector`, `DataQualitySummary`, `ImportWizard`, `ValidationIssuesTable`, `SourceComparisonTable`, `TwoByTwoTable`, `StatisticValue`, `EvidenceStatus`, `TimelineControl`, `SourceDetail`, `ReportDownload`, and optional `AiDraftPanel`/`ResponsePlanner`. Display numerator/denominator; use “Not estimable” with reason instead of blank cells. Charts require equivalent textual/table summaries and labels. Color alone never conveys status.

Disable duplicate submit while requests run, but keep server-side idempotency. Confirmations must state row inclusion/exclusion and warning counts. Stale snapshots are visibly labelled after a new import. Preserve form data on safe retry. Destructive actions require confirmation; deletion policy remains TBD.

14. UI/UX and Design System

14.1 Approved palette

The following are the only permitted base-colour tokens. No additional hexadecimal, RGB, named accent, or status colours may be introduced. Status meaning must also be conveyed by text, icon shape, border/pattern, and accessible labels; chart categories need direct labels or legends.

Token	Value	Use
`--color-ink`	`#111827`	Main headings, primary text, navigation emphasis, dark surfaces
`--color-slate`	`#374151`	Secondary text, borders, supporting controls
`--color-mist`	`#F3F4F6`	Page background, neutral grouping and hover surfaces
`--color-white`	`#FFFFFF`	Main content surfaces and text on dark surfaces

Recommended approved pairings: ink on white or mist; slate on white or mist for non-small secondary content only after contrast calculation; white on ink or slate. Do not use slate as small text on mist unless measured contrast is at least 4.5:1. Ink on mist is approximately 13.3:1; ink on white approximately 17.7:1; white on ink approximately 17.7:1; white on slate approximately 10.3:1. All actual text, focus, border, and chart combinations must be contrast-tested. Do not rely on colour as the sole signal. Numeric contrast ratios should be rechecked by an automated accessibility test against the final CSS.

14.2 Typography and global styling

Only two primary typography roles are defined: **Heading** and **Body**. A single supported sans-serif family with weight/size variants is preferred; reuse the existing project family if present. Exact font family and final type scale are TBD until repository inspection/design review. Headings have a consistent page/section/component hierarchy. Body covers descriptions, labels, forms, tables, statistical explanations and limitations with readable line-height and tabular numerals for aligned statistics.

A mandatory global stylesheet (`app/globals.css` or repository-equivalent) defines the four custom properties, heading/body styles, spacing scale, content widths, border/focus styles, sizing, page background, buttons, inputs, links, cards, tables and dialogs. Shared components consume the tokens; avoid arbitrary page-specific colours and CSS. Document any justified exception. Honor `prefers-reduced-motion`; use motion only when it communicates state and never as a prerequisite to understanding.

14.3 Visual and responsive behaviour

The visual language is a restrained investigation workstation: evidence-first hierarchy, compact but readable tables, visible provenance, aligned counts and rates, consistent navigation and explicit quality states. Prohibited patterns include AI gradients, glassmorphism, decorative fake charts, repeated low-value cards, excessive rounded surfaces/shadows, emoji icon systems and dashboard-wide chatbot placement.

Desktop is the primary work surface. At narrow widths, navigation collapses accessibly; forms stack; source comparisons use controlled horizontal overflow or a purposeful stacked row presentation; important row/source labels remain discoverable; charts resize with a text/table alternative; report downloads remain reachable. Breakpoint values are an implementation configuration to document after checking the installed framework/design system. Tables must not cause whole-page horizontal overflow; only their labelled container may scroll.

14.4 Accessibility target

Target WCAG 2.2 Level AA. Use semantic landmarks/headings, labelled inputs, accessible names, keyboard-operable controls, visible focus, logical focus order, accessible dialogs, announced validation/status changes, and meaningful error text. All actions work without a pointer. Charts need summaries and an equivalent data table. Status must not be conveyed by colour alone. Test with automated accessibility checks plus keyboard-only review and manual screen-reader sampling. A successful automated scan alone is not proof of conformance.

14.5 Design-system requirement register

ID	Priority	Required behaviour	Verification
UI-001	MUST	Use only the four approved base palette tokens; no fifth base colour or colour-only status	CSS scan + visual review
UI-002	MUST	Central global stylesheet defines tokens, spacing, borders, focus, sizing and reusable control defaults; exactly two primary type roles, Heading and Body	Stylesheet inspection + component consistency review
UI-003	MUST	Every async workflow has usable loading, empty, success and error states; chart values have textual/table alternatives	Component/E2E/accessibility tests
UI-004	MUST	Desktop/tablet/mobile layouts preserve labels and core actions without page-wide horizontal overflow	Responsive browser tests at agreed viewports
UI-005	MUST	Target WCAG 2.2 AA with semantic HTML, keyboard access, focus, labels, accessible validation and non-colour-only status	Automated scan plus manual keyboard/screen-reader review

15. Data Requirements and CSV Import Contract

15.1 Canonical person-level CSV contract

The import contract below is the proposed MVP contract; adjust only by a documented schema-version change. Headers are UTF-8 text, case-sensitive after trimming surrounding whitespace; duplicate headers after trimming are a blocking error. Accept UTF-8 and UTF-8-with-BOM. Other encodings are not auto-guessed;

the user must convert/re-upload. Delimiter is comma. Quoted fields follow RFC 4180-compatible CSV parsing; malformed quoting is fatal for the file because row boundaries cannot be trusted. Do not execute formulas or interpret cells as code.

Header	Required	Accepted form	Rule
`participant_id`	Yes	Nonempty string, max 128 chars	Unique within one imported version after whitespace trim; duplicate rows are rejected/ held for review, never silently merged
`illness_status`	Yes	`ill`, `not_ill`, `unknown`	Case-normalise only for documented enum matching; other values are row errors
`recorded_at`	Yes	ISO 8601 datetime with explicit offset or `Z`	Normalise to UTC; invalid or timezone-naive timestamps reject that row
`exposure__<source_key>`	At least one	`exposed`, `unexposed`, `unknown`	One column per declared source key; keys match `^[a-z0-9][a-z0-9_-]{0,63}$`
`survey_date`	No	`YYYY-MM-DD`	Invalid provided date is a row warning and stored as null; never reinterpret a date as local time
`symptom_onset_date`	No	`YYYY-MM-DD`	Invalid provided date is a row warning and stored as null; onset after recorded-at date may be flagged for review, not silently corrected
`household_group_id`	No	Pseudonymous string, max 128 chars	Do not require address or identifiable household details
`record_provenance`	No	Bounded plain text, max 500 chars	Treat as untrusted text; do not permit it to control prompts or HTML

A source key is an opaque stable identifier, not an arbitrary display name. Source display names/types and optional location metadata are created/maintained as source metadata within the investigation. Import may create an explicitly reviewed source mapping during preview; no column may cause the system to invent source coordinates or provenance.

Recommended safety limits for the MVP, subject to measurement/approval (DEC-03): 30 MiB uploaded CSV, 100,000 data rows, 100 columns, and 20 sources per import. Enforce limits server-side before expensive parsing and during parsing; a file over any limit fails without partial import. A configurable file limit must never exceed the tested memory envelope. Reject NUL bytes and malformed UTF-8; do not infer that a blank exposure value means unexposed.

15.2 Row treatment and import transaction

Blocking file errors: unsupported file type/encoding, over-limit size/rows/columns/sources, malformed CSV, missing required header(s), duplicate headers, absent source exposure columns, and schema ambiguity that prevents safe interpretation. Such a file cannot be committed.

Row errors: missing/duplicate participant ID, invalid illness enum, invalid/missing `recorded_at`, or invalid exposure enum. The preview identifies the exact row, column, code and human-readable remediation. Default proposed policy is **reject invalid rows from the import, preserve them in the validation report, and require an explicit review/confirmation showing exact accepted/rejected counts**; the raw upload is not silently rewritten. The import records the contract/schema version, SHA-256 of the exact uploaded byte stream (lowercase 64-character hex), and validation report. This fingerprint binds preview to the exact submitted file; it is not anonymisation. If all rows are rejected, do not create a usable dataset version.

Non-blocking warnings: unknown illness status, explicit unknown exposures, absent optional onset/survey dates, invalid optional date (stored null and disclosed), possible date-order inconsistencies, and source metadata gaps. Unknown illness rows remain imported for data-quality totals but are excluded from source-specific 2×2 illness comparisons. Unknown exposure is counted separately for that source and excluded from both exposed and unexposed denominators. A missing required exposure value is a row error, not an implicit unexposed classification. Warning acknowledgements are recorded at import confirmation.

Import validation returns a preview/report with original row number, field, rule code, severity, safe message, and action (`fix_and_reupload`, `acknowledge`, `rejected`). A rejected-row CSV includes original values with safe CSV quoting and a validation reason; protect against spreadsheet-formula injection in exported text by escaping formula-triggering cells. Do not log raw rows.

15.3 Provenance and synthetic/reference separation

Every import/version records provenance kind: `synthetic_demo`, `field_survey`, `published_aggregate_reference`, or `other_user_supplied`. A person-level participant import must not be used to represent aggregate-only published evidence. Reference tables are separately typed and labeled, including source title/URL/publication context and the data limitation. Synthetic source locations must carry a `synthetic` designation and visible UI/report label. Dataset content, origin, upload timestamp, hash, schema version, accepted/rejected/warned counts, transformations and acknowledgements are traceable.

15.4 Data retention minimisation

Participant IDs must be pseudonymous. Names, phone numbers, government IDs, full addresses, exact household coordinates and other direct identifiers are not required and must be rejected or removed through an explicit controlled data minimisation policy—not silently discarded. Exact record storage/deletion periods and backups retention require approval (DEC-06). Retention defaults must not be presented as policy until approved.

15.5 Data requirement register

ID	Priority	Required behaviour	Verification
DATA-001	MUST	Enforce versioned CSV headers, enums, timestamp/ date formats, source-key syntax and approved resource limits server-side	Parser/schema tests
DATA-002	MUST	Record received/accepted/ rejected/warned accounting and report each exclusion/ transformation; never silently change source rows	Import reconciliation tests
DATA-003	MUST	Preserve unknown illness/ exposure distinctly; unknown exposure never enters either known exposure group	Unit/API/UI tests
DATA-004	MUST	Each committed version is immutable and tied to upload-byte SHA-256, schema version, provenance and validation summary	Import/version integration tests
DATA-005	MUST	Source coordinate pairs are range-validated, provenance-tagged and never inferred; synthetic points are labelled	Source API/map/report tests
DATA-006	MUST	Participant data use pseudonymous IDs and are not exposed in public pages, URLs, ordinary logs or external LLM payloads	Security/privacy tests

16. Epidemiological Methodology and Statistical Rules

16.1 Source-specific table and denominators

For each source, construct the 2×2 table from rows that have both a known illness status and known exposure status for that source:

	Ill	Not ill
Exposed	a	b
Unexposed	c	d

`unknown` illness is neither `ill` nor `not_ill`; `unknown` exposure is neither `exposed` nor `unexposed`. For each source, report two separate breakdowns: (1) unknown-exposure records grouped by illness status (`ill`, `not_ill`, `unknown`), and (2) unknown-illness records grouped by exposure status (`exposed`, `unexposed`, `unknown`). Also report the intersection count where both illness and exposure are unknown, and the unique number of records with at least one of those statuses unknown. A record in the intersection appears in both breakdowns; do not add the breakdown totals together. Participants may be exposed to multiple sources; analyse each source independently, without assigning participants to mutually exclusive source categories. Thus source-specific marginal counts can overlap and must not be summed as if they were partitions.

16.2 Deterministic measures

- Exposed attack rate: `a / (a + b)`.
- Unexposed attack rate: `c / (c + d)`.
- Relative risk (RR): `[a / (a + b)] / [c / (c + d)]`.

Show each rate as a percentage plus its numerator and denominator. Never present an unrounded value alone as a precise finding. Preserve full precision internally; round for display only using one documented formatting helper. If either known group denominator is zero, its attack rate is unavailable; RR is unavailable if either attack rate is unavailable or the unexposed attack rate is zero. Return `null` for unavailable values plus a machine-readable reason (for example `ZERO_UNEXPOSED_CASES`, `EMPTY_EXPOSED_GROUP`, `EMPTY_UNEXPOSED_GROUP`); never serialize `NaN` or `Infinity`.

16.3 Proposed confidence-interval method

Recommended default pending approval (DEC-04): Katz log confidence interval for the RR, a deterministic log-Wald interval. When `a > 0`, `c > 0`, and both exposed/unexposed denominators are positive:

`SE(log(RR)) = sqrt(1/a - 1/(a+b) + 1/c - 1/(c+d))`

`95% CI = exp(log(RR) ± 1.96 × SE(log(RR)))`.

No continuity correction is applied. If required cells are zero or the SE is undefined, return no CI with reason `CI_UNAVAILABLE_ZERO_CASE_CELL`; do not invent a bound. If any 2×2 cell is below 5, attach `SMALL_CELL_APPROXIMATION` and explain that the asymptotic interval may be unreliable. This is not a universal preferred method for every dataset; the method must be approved by a qualified epidemiology reviewer before operational use. Unit tests must use independently calculated expected values.

16.4 Proposed evidence sufficiency and candidate status

The following is a transparent **proposed rule**, not an epidemiological diagnosis. The minimum known sample count of 10 in each exposure group is a provisional guardrail only (DEC-05) and requires epidemiology review.

1. If either group has zero denominator or fewer than 10 records with known illness and exposure, status is `INSUFFICIENT_DATA_TO_COMPARE`.

2. Otherwise, calculate RR and the approved CI deterministically.
3. If RR > 1 and a valid 95% CI lower bound > 1, status is `HIGHER_PRIORITY_CANDIDATE`.
4. If RR < 1 and a valid 95% CI upper bound < 1, status is `LOWER_OBSERVED_ASSOCIATION`.
5. Otherwise status is `INCONCLUSIVE_EVIDENCE`.
6. Add `REQUIRES_ADDITIONAL_VERIFICATION` as an independent flag when there are material missing/unknown values, small-cell warnings, missing source metadata, low source-specific coverage, inconsistent data, or absent confirmatory lab/source inspection evidence. The application must state the actual reason(s); do not use a hidden composite score.

Sort candidates deterministically by status order (`HIGHER_PRIORITY_CANDIDATE`, `INCONCLUSIVE_EVIDENCE`, `LOWER_OBSERVED_ASSOCIATION`, `INSUFFICIENT_DATA_TO_COMPARE`), then by RR descending only where both RRs are estimable, then by stable `source_id` ascending. Label this as a display ordering rule, not proof of causal priority. If statuses tie, show the tie explicitly; source ID is only a reproducibility tie-breaker. If CI unavailable, do not award a higher-priority status based on RR alone. Priority/ranking never says “confirmed contaminated.”

16.5 Synthetic fixture acceptance targets

These are required synthetic acceptance targets from the supplied specification; repository fixture existence and execution are **not verified**. Counts below are source-specific marginals. `unknown exposure` means known illness but exposure not classifiable for that particular source; illness-unknown rows are tracked separately.

Source	Exposed ill/not ill (a/b)	Unexposed ill/not ill (c/d)	Unknown exposure ill/not ill	Expected RR	Exposed AR	Unexposed AR
Handpump 14	42 / 28	24 / 136	6 / 4	4.0	60%	15%
Well 3	30 / 70	36 / 84	6 / 14	1.0	30%	30%
Community Tank 2	24 / 56	36 / 84	12 / 28	1.0	30%	30%

The fixture-level cohort target is 240 participants, 72 ill and 168 not ill. The source marginals above must match exactly; do not require the source groups to sum to 240, because exposure overlaps are valid. Tests must separately verify global illness totals and each per-source table.

16.6 Published aggregate benchmark: CDC Sancaktepe Village

The cited CDC Field Epidemiology Manual example reports the following **aggregate** table:

Exposure group	Ill	Well	Total	Attack rate
Exposed to tap water	46	127	173	26.6%
Unexposed	9	76	85	10.6%

The reported RR is approximately 2.5, with published 95% CI 1.3–4.9. This is an aggregate calculation example, not participant-level raw data. The CDC example concerns oropharyngeal tularemia in Sancaktepe Village; it must not be relabelled as cholera or represented as a person-level import. See Reference [1].

16.7 Published aggregate daily counts: Mehar outbreak

This published reference is a daily aggregate case-count time series associated with a reported cholera outbreak in Mehar village, Madhya Pradesh, in July 2024. Keep the exact values/date labels and do not create a 2 July value.

Date	HWC Mehar	Base Camp	DH Sagar	BMC	Total
2024-07-03	0	0	0	1	1
2024-07-04	140	0	15	5	160

Date	HWC Mehar	Base Camp	DH Sagar	BMC	Total
2024-07-05	220	71	41	3	335
2024-07-06	58	74	19	3	154
2024-07-07	73	55	9	2	139
2024-07-08	17	34	6	0	57
2024-07-09	22	11	3	0	36
2024-07-10	11	7	1	0	19

The supplied reference total is 901, with peak daily total 335 on 5 July 2024. These values can test an aggregate timeline chart and total checks only. They cannot validate participant-level source exposure or RR. See Reference [2].

16.8 Timeline replay correctness

At selected UTC cutoff `T`, the eligible participant set is exactly `recorded_at <= T`; all source-specific counts, denominators, unknown indicators, CI, status and sort order are recomputed from that eligible set. Do not filter only the visible chart or reuse a later full-dataset analysis. Record `T`, dataset version, inclusion rules and statistical-method version in analysis metadata and reports. `survey_date` and `symptom_onset_date` represent different concepts; neither substitutes for `recorded_at`. If no rows qualify, return a valid empty-state result rather than a prior analysis. Timestamp display is localized only in presentation, with UTC stored/compared consistently and precision documented.

16.9 Statistical requirement register

ID	Priority	Required behaviour	Verification
STAT-001	MUST	Build the source-specific 2×2 table from rows with known illness and known exposure for that source; count unknowns separately	Independent table unit tests
STAT-002	MUST	Calculate exposed/unexposed attack rates and RR from documented numerators/ denominators; display both numerator and denominator	Independent formula tests + API/UI tests
STAT-003	MUST	Return null plus reason for undefined rates/RR/CI; never emit NaN or Infinity	Zero-denominator/serialization tests
STAT-004	MUST	Use a versioned, approved 95% RR CI method and deterministic zero-cell/small-sample handling	Independent CI edge-case tests; method approval record
STAT-005	MUST	Apply explicit deterministic sufficiency, status, tie-break and verification-flag rules; LLM cannot change them	Ranking unit/property tests
STAT-006	MUST	Analyse overlapping source exposures independently and disclose non-causal/ confounding limitations	Multi-exposure tests + report review
STAT-007	MUST	Timeline includes only `recorded_at <= T` in UTC and recomputes every statistic, status and order for the eligible data	Future-record leakage regression tests

17. Backend and API Requirements

The proposed API contract is detailed in `AquaTrace_API_Specification.md`; that document is authoritative only after implementation review and must remain consistent with this SRS. The backend owns request validation, import contracts, access control, dataset versioning, deterministic calculations, timeline filtering, source metadata, safe AI calls, reports, error mapping and audit logging. OpenAPI generated by the installed FastAPI version must match contract schemas; examples below are illustrative contracts, not claims of an existing API.

17.1 Common API rules

- JSON requests/responses use UTF-8. Upload endpoints use `multipart/form-data`; export endpoints return a bounded streaming/download response.
- All investigation-scoped routes enforce access to the investigation and all referenced dataset versions on the server.
- A dataset version is immutable. An analysis request names an explicit version or the server returns a versioned result that records exactly what was analysed. Never silently analyse the “latest” version after user selection changed.
- Use standard HTTP status codes and one consistent error envelope: `{ "error": { "code": "VALIDATION_FAILED", "message": "The request could not be accepted.", "request_id": "...", "details": [{"field":"...","code":"...","message":"..."}] } }`. Do not expose stack traces, MongoDB errors, secrets, raw rows or identifiers in error messages.
- Use request IDs in responses/logs. Upload commit should support an idempotency key or equivalent server-side duplicate-submission protection. Retryable client errors must explain whether commit occurred.
- Private investigation responses, exports, AI output and participant-level data are not publicly cached. Auth method/session mechanics are TBD pending repository/security review.

17.2 Candidate route catalogue

Required MVP routes are proposed in the API Specification: health/readiness; create/list/read/update investigations; validate/preview CSV; commit an approved import; list/read immutable dataset versions; run/read analysis; source comparison/detail; timeline replay; report generation/download. Source metadata is investigation-scoped. AI summary is optional and failure-isolated. Task/inventory routes are optional only after the core workflow meets acceptance criteria. This list is not evidence that these routes already exist.

18. Database Requirements

MongoDB is the selected database. The proposed document model in `AquaTrace_Data_Dictionary.md` uses a small number of collections: investigations, dataset versions/import metadata, participant records tied to an immutable dataset version, source metadata, analysis snapshots/metadata, report metadata, and an audit collection where access/change events are required. Optional task/resource entities are added only when the Response Planner is approved.

Dataset participant records are scoped by both investigation and immutable dataset version. Import commit must either create the complete version and its records atomically or leave no usable partial version; use a transaction where supported by the selected MongoDB topology, otherwise an explicit staging/commit marker and recovery process. An incomplete staged import is never selectable as an analysis version. Do not overwrite imported rows in place. Corrections arrive as a new version with provenance. Analysis snapshots capture dataset version hash, cutoff, filters, calculation-method version, CI method version and relevant source metadata version. Avoid storing redundant derived statistics unless needed for reproducible snapshot/report performance; if persisted, treat them as immutable computed artifacts with a versioned key.

Recommended indexes include unique `(investigation_id, version_id)` on dataset metadata; `(investigation_id, dataset_version_id, participant_id)` unique among imported rows; `(investigation_id, dataset_version_id, recorded_at)` for timeline filtering; `(investigation_id, source_id)` unique for source metadata; and index fields used for analysis snapshot keys. Indexes must be confirmed against actual query plans and representative data. Database schema validation should enforce types/enums and bounded fields where practical. Deletion, backup retention, record-level access, and audit retention remain subject to DEC-06/DEC-07.

19. LLM, LangChain, LangGraph, and Groq Requirements

19.1 Scope and authority

AquaTrace calls an existing, explicitly configured Groq-hosted LLM; no model training, fine-tuning pipeline, training dataset, training GPUs or training budget are in scope. LangChain/LangGraph are used only for the bounded, optional AI explanation/draft workflow. Do not add multiple agents merely to advertise a framework.

Deterministic Python functions are the sole authority for epidemiological counts, attack rates, RR, CI, warnings derived from data rules, candidate statuses, tie-breaking, and task/resource constraints. The LLM may explain already computed results, summarise explicitly supplied limitations, organise user-reviewed notes, and suggest protocol-based questions/tasks for human review. It may not alter statistics, add findings, infer missing records, invent sources/citations/guidelines, decide contamination, issue treatment/emergency orders, or assign resources.

19.2 Context sent to provider

Backend constructs a strict allowlisted, versioned JSON object. Example contract (values are illustrative placeholders, not a real result):


```
{
  "schema_version": "1.0",
  "investigation": {"title": "User-supplied non-identifying title", "area_label": "User-supplied coarse area label"},
  "dataset": {"version_id": "immutable-version-id", "provenance_kind": "synthetic_demo", "included_records": 0,
        "excluded_records": 0, "recorded_through_utc": "2026-01-01T00:00:00Z"},
  "analysis": {"method_version": "approved-method-id", "source_results": [{"source_id": "source-key", "source_name":
        "source label", "a": 0, "b": 0, "c": 0, "d": 0, "attack_rate_exposed": null, "attack_rate_unexposed": null,
        "relative_risk": null, "rr_ci_95": null, "priority_status": "INSUFFICIENT_DATA_TO_COMPARE", "limitations": ["No
        estimable comparison"]}]},
  "reviewed_field_notes": [],
  "allowed_output_purpose": "draft_explanation_for_authorised_human_review"
}
```


Only send aggregate outputs and the minimum field-note text needed. Never send participant rows, participant/household IDs, direct identifiers, raw uploaded files or exact household coordinates. Free-text notes are untrusted. Apply explicit prompt boundaries and delimit notes as data, not instructions. Provider terms/data-retention behavior must be verified against the configured account/policy before real-health use (DEC-08); do not claim a no-retention guarantee without evidence.

19.3 Structured output and validation

Proposed response schema: `schema_version`, `summary` (bounded plain text), `source_explanations[]` each with `source_id`, `evidence_statement`, `limitations[]`, `follow_up_questions[]`, and `review_required: true`. Only source IDs present in the request are allowed. Each statement must be supported by the allowlisted statistics/limitations; no citations unless independently provided and verified by the application. Enforce output type, maximum length, array limits, UTF-8 and text escaping. Reject unknown fields or IDs, unsupported numerical claims, medical treatment instructions, unsubstantiated source-causation claims, and attempts to override the allowed task. The precise JSON schema should be versioned in code and the prompt version recorded with the draft.

19.4 LangGraph operation and safe fallback

One bounded graph is sufficient: build allowlisted context → call configured model with timeout → validate structured output and claims against context → either return a draft for human review or reject it → present explicit fallback if unavailable. Proposed initial timeout: 15 seconds per provider request and 20 seconds overall for the graph. Maximum retries: one retry for transient/rate-limited failures only, and only if the overall deadline allows it; do not retry authentication or invalid schema endlessly. These timeouts are configuration defaults to measure/approve, not current provider performance claims. Configurable model identifier must be explicitly set; exact supported model ID is TBD. API key is read only by the server from a non-`NEXT_PUBLIC_` environment variable. Do not log prompts/raw notes. Save only validated draft metadata/output and provenance if policy allows; graph-state persistence/checkpointing is not mandatory for the MVP. Graph invocation, prompts and model configuration need versions to reproduce behavior.

Missing key, timeout, rate limit, provider outage, malformed output or safety validation failure must not block deterministic analysis, ranking, timeline or report generation. UI shows “AI explanation unavailable” or “Draft rejected; use deterministic results” without claiming the draft was created. The deterministic report can be generated without the AI panel.

19.5 AI requirement register

ID	Priority	Required behaviour	Verification
AI-001	MUST when AI is enabled	Backend sends only versioned allowlisted aggregates, approved metadata and reviewed notes; never participant rows/direct identifiers	Payload inspection + privacy tests
AI-002	MUST when AI is enabled	Enforce structured output schema, allowed source IDs, bounds and evidence-grounded claims; invalid drafts are rejected	Schema/adversarial output tests
AI-003	MUST	Groq/model selection and key remain server-side; missing key/outage/malformed output cannot block deterministic features	Secret scan + fault-injection E2E
AI-004	MUST when AI is enabled	Imported free text is untrusted; model cannot change statistics/statuses or execute unrelated tasks; draft requires human review	Prompt-injection and authority-boundary tests

20. Caching and Performance Requirements

20.1 Cache policy

Correctness, privacy and version identity take precedence over cache hit rate. Never place private investigation data, participant rows, report downloads or LLM prompts in public/shared caches. Authenticated investigation API responses should default to `Cache-Control: private, no-store` unless a reviewed route-specific policy proves otherwise. Static JS/CSS/font assets may use immutable caching when fingerprinted. Stable non-sensitive reference metadata can use a bounded cache with explicit version and TTL.

Backend analysis cache key must include at least: investigation ID, immutable dataset version ID and fingerprint, canonical filter/inclusion rules version, UTC timeline cutoff (or explicit “full version”), statistical-method/CI version, source metadata version where it affects output, and output schema version. The exact canonical key algorithm must be stable and tested. Dataset version changes cannot reuse an older version’s cached analysis. Missing cache is a normal cache miss; cache outage falls back to deterministic computation if safe, not an incorrect cached result.

Invalidate or bypass cached results when a dataset version is committed, source metadata relevant to output changes, calculation method changes, filters change, or a report is requested for a different snapshot. An analysis/report response carries the snapshot identifiers it represents. UI compares these identifiers with the selected dataset and displays a stale-version warning or refetches; never label an older result “latest”. Proposed initial in-process cache defaults (not benchmarked): analysis snapshots TTL 15 minutes with LRU eviction capped at 1,000 entries or 256 MiB per worker, whichever is reached first; investigation-scoped source metadata TTL 5 minutes; versioned public reference metadata TTL 5 minutes. Cache only validated derived snapshots/metadata, not raw participant rows or prompts. Eviction and restart are normal cache misses. These limits must be configurable and re-sized only after memory/performance measurement (DEC-09). Do not cache raw participant records or prompts by default. Any deviation needs privacy justification and approval.

Frontend behavior must follow the installed Next.js version: fingerprinted static assets may use `public, max-age=31536000, immutable`; stable non-sensitive public reference metadata may use a 5-minute TTL. Investigation-specific HTTP responses remain private/no-store. Any permitted client-side in-memory reuse is scoped by authenticated session + investigation/version and bounded to 30 seconds; purge it on logout and invalidate it after upload, edits, deletion or analysis configuration changes. Do not persist sensitive payloads in browser local storage. Do not assume a particular Next.js cache API without inspecting installed versions.

20.2 Lazy loading

Lazy-load map libraries and source maps, optional charts not needed in the first viewport, long history views, report previews, AI explanation panel and optional Response Planner screens. Use pagination or virtualisation for large tables where measured rendering justifies it. Keep primary navigation, investigation identity, provenance, key counts and next-step actions available immediately; don’t lazy-load critical first-screen information. Every lazy boundary has loading, empty and failure states, with a usable table/text fallback. Provider/map failure cannot block analysis/reporting.

20.3 Proposed measurable targets

These are proposed hackathon targets, not measured performance claims. Verify in a documented reference environment with version, hardware, browser, warm/cold-cache state and dataset fixture. Revisit after first measurement (DEC-09).

ID	Priority	Proposed target / required behaviour	Verification method	Measurement conditions
PERF-001	SHOULD	Dashboard primary summary usable within 2.5 s at p95	Browser performance trace	Production build, cold navigation, agreed reference laptop/ network
PERF-002	SHOULD	CSV preview/validation ≤5 s for 10,000 rows / 5 MiB benchmark fixture	Timed upload/validation test + peak-memory measurement	Parser only; no database commit
PERF-003	SHOULD	Deterministic analysis for 100,000 accepted rows and 20 sources ≤3 s p95	Load/benchmark script	API entry to validated response on named hardware
PERF-004	SHOULD	Exact-key cached timeline snapshot ≤500 ms p95	Cache-hit benchmark	Same full immutable key; cache state recorded
PERF-005	SHOULD	Uncached timeline replay over 100,000 rows / 20 sources ≤3 s p95	Recompute benchmark	Full recomputation, not visual-only filtering
PERF-006	SHOULD	Field brief for a 100,000-row version ≤5 s p95	Report-generation benchmark	Aggregated report; no participant-row dump

ID	Priority	Proposed target / required behaviour	Verification method	Measurement conditions
PERF-007	MUST	Over-limit input rejected before unbounded allocation; parsing stays within documented service memory/CPU budget	Adversarial limit test + memory/time profile	Boundary, malformed and oversized inputs
PERF-008	MUST	Map, optional charts, report preview and AI panel are lazy-loaded; critical navigation/ summary remains eager	Bundle analysis + browser performance trace	Inspect initial route and load optional modules only on demand

Use representative and adverse inputs, not only empty or tiny fixtures. If a target fails, report the measurement and mitigation; do not change the target to hide an unrecorded failure. Hardware and deployment topology remain TBD.

21. Lazy Loading and Responsive Behaviour

Lazy loading must be applied to optional/heavy modules as described in Section 20. A loading fallback must announce activity where needed, reserve layout space to reduce content shift, and provide an error state with a retry or non-interactive fallback. Dynamic imports must not hide runtime errors. Browser zoom to 200% must retain access to core workflows; mobile/tablet layouts must preserve evidence labels and avoid page-wide horizontal scrolling. A source comparison table can use a labelled horizontal scroll region where a meaningful stacked view is not practical. Keyboard focus stays logical after route changes, modal closure and asynchronous updates.

22. Security, Privacy, and Audit Requirements

22.1 MVP security baseline

ID	Priority	Control	MVP requirement / verification
SEC-001	MUST	Secret management	API keys and DB credentials are server-side environment/ secret-manager values; never committed, shipped to browser, or written to logs. Search built assets and repository config.
SEC-002	MUST	Controlled access	Before real data are accepted, authentication and investigation-level authorisation must be configured and tested. For a public hackathon demo, use synthetic-only data or a documented restricted-access deployment; public write access is not acceptable.
SEC-003	MUST	Tenant/investigation isolation	Every scoped read/write checks investigation access in backend; test cross-investigation IDs, guessed IDs and exports.

ID	Priority	Control	MVP requirement / verification
SEC-004	MUST	Upload security	Enforce content/extension policy, size/row/column limits, safe CSV parsing, row/column validation, timeouts and rate limits; do not trust MIME type alone.
SEC-005	MUST	Injection / output safety	Escape imported text in HTML and reports; never interpolate it into SQL-like commands or model/system instructions. Protect CSV exports from formula injection.
SEC-006	MUST	Browser protections	Configure HTTPS in deployed environments, secure cookies if cookie sessions are chosen, appropriate CSP/security headers and CSRF controls where cookie-authenticated state changes are used. CORS is explicit allowlist, not wildcard with credentials.
SEC-007	MUST	Query safety	Use parameterized/typed Mongo operations, allowlisted sort/filter fields and bounded pagination; never allow user-supplied arbitrary query operators.
SEC-008	MUST	Export protection	Authorise before generation/ download; use safe content disposition/filename, private storage/headers and short-lived authenticated downloads if files are stored.
SEC-009	MUST	Rate/resource limits	Bound upload, timeline, report and AI routes; use request deadlines and concurrency controls suitable for deployment.
SEC-010	MUST	Logging	Structured event logs with request ID, actor pseudonymous ID where available, investigation/version IDs and action outcome; no raw participant rows, tokens, secrets or full prompts.
SEC-011	MUST	Dependencies/config	Pin reviewed dependency ranges/lockfiles where repository conventions support it; inspect known vulnerabilities and use separate development/ production config.
SEC-012	MUST	Audit trail	Record dataset import/commit, metadata edits, export generation, task/inventory changes and deletion approvals with actor/time/ action/target; avoid putting sensitive field values in audit events.

This document does not certify production readiness. The repository could not be inspected, so actual controls are unverified. Authentication scheme, hosting, threat model review and access policy remain open decisions. A prototype using synthetic data may use a controlled demo gate, but that is not an adequate production control for real health records.

22.2 Privacy and retention

Use pseudonymous participant IDs, coarse geographic area labels and source-level coordinates only when trustworthy and necessary. Exact participant/household coordinates are outside the MVP. Do not make participant-level health/exposure data public, encode it in URLs, expose it through logs, include it in browser error traces, or send it to Groq. Separate synthetic demonstration data from all real datasets using provenance labels and visibly distinct environment/data-set designation, not colour alone.

Access, correction, export and deletion rights must be explicitly assigned. Retention duration, deletion workflow, backups and audit retention require approval (DEC-06); do not invent a statutory or organisational retention period. Delete/revoke stored reports or ensure they follow dataset/investigation access and deletion policy. A hash may be stored for integrity/version identity; it is not anonymisation. Review whether any combination of coarse metadata could identify a person in a small community.

23. Reliability and Failure Handling

ID	Priority	Failure scenario	Required behaviour / verification
NFR-001	MUST	MongoDB unavailable	Return safe 503/service-unavailable envelope; do not claim imports/edits committed. Health endpoint separates process liveness from database readiness.
NFR-002	MUST	Malformed/oversized CSV	Reject with actionable code/ row report where safe; no partial committed version; bound CPU/memory.
NFR-003	MUST	Row-level errors	Show rejected/warned counts and exact remediation; commit only the expressly confirmed accepted set; persist validation report and fingerprint.
NFR-004	MUST	Interrupted upload/duplicate retry	Expire temporary staging safely; use idempotency/ deduplication; report whether commit happened.
NFR-005	MUST	Analysis for nonexistent/wrong investigation version	Return 404/422 as appropriate; never silently switch to latest.
NFR-006	MUST	Groq missing/outage/invalid result	Return deterministic results normally; omit AI draft with explicit unavailable/rejected status.
NFR-007	MUST	Map provider unavailable/ coordinates absent	Hide or disable map layer with explanation; source table and source detail remain available.
NFR-008	MUST	Report generation failure	Return controlled error, do not expose a partial/truncated file as success; provide retry while preserving selected snapshot.
NFR-009	MUST	Cache miss/outage/stale result	Recompute or fail safely; never use a key that omits version/ method/cutoff.
NFR-010	MUST	Empty investigation/dataset	Clear empty state with next action; do not show placeholder counts/charts as data.

ID	Priority	Failure scenario	Required behaviour / verification
NFR-011	MUST	Insufficient groups / zero cells	Return nullable statistics plus reason/limitations; no NaN/ Infinity and no unsupported priority promotion.
NFR-012	MUST	Frontend request aborted	Show cancellable/in-progress/ retry status based on server-confirmed state; do not imply server transaction rolled back without confirmation.
NFR-013	MUST	Invalid/missing map coordinates	Exclude from map, state reason; do not geocode or infer a point silently.
NFR-014	MUST	Private investigation data cache policy	Participant-level data, reports and AI prompts use private/no-store response policies; verify headers and CDN/reverse-proxy behavior.
NFR-015	MUST	Analysis cache key/invalidation	Key includes investigation, immutable dataset version/ fingerprint, cutoff, filter/ inclusion version, calculation/CI version, relevant source metadata version and output schema; prove every dimension prevents stale reuse.

Unhandled backend exceptions are logged securely with request IDs and translated to a generic 500 response. Expose diagnostic details only in controlled development mode, never for real health data or public deployments. Readiness probes reflect dependencies required to serve traffic; optional Groq/map outages should not mark core deterministic analysis unavailable.

24. PDF Reporting Requirements

The field brief is an evidence-linked snapshot, not an autonomous recommendation. Its header/footer include investigation title and stable ID, generation UTC timestamp, selected dataset version/hash or short fingerprint, selected timeline cutoff, analysis-method version, provenance designation and page numbering. Include:

1. Short summary with evidence status and explicit non-diagnostic context.
2. Dataset row accounting: received, accepted, rejected, included, excluded and flagged counts, with definitions.
3. Candidate source table with display order, source status, exposed/unexposed `a,b,c,d`, group denominators, attack rates, RR, 95% CI where estimable, unknown exposure/illness counts and warnings.
4. Data-quality/provenance and limitations section, including synthetic vs aggregate reference vs field-survey designation.
5. Timeline snapshot time and the record-availability cutoff rule when replay is selected.
6. Alternative explanations/confounding, co-exposures and limits of source-by-source marginal comparisons.
7. Suggested source inspection, sampling/verification or follow-up steps as review prompts for the authorised officer, never automatic orders.
8. Optional task checklist/status only if planner is enabled and tasks were in the selected snapshot.
9. Methods/version and disclaimer: **an observed statistical association does not establish that a source is contaminated or prove causation**.

All report numbers/statuses are rendered from the same immutable analysis snapshot used by the screen; the LLM is not the source of report statistics. AI prose, when included, is visibly labelled as a human-review draft, independently validated, and cannot change the numeric tables. Report generation validates snapshot ownership/current user permissions at both generation and download. Escape untrusted text, avoid participant-row exports and keep report files private. PDF or print-ready implementation choice is TBD until dependencies and deployment are inspected. A report should paginate tables, repeat table headers, keep text selectable/searchable where feasible and avoid clipped columns/figures.

25. Optional Response Planner Requirements

The Response Planner is an optional extension and cannot delay the core CSV → validation → deterministic analysis → timeline → brief workflow. MVP priority is checklist/task management only if sufficient time remains; resource inventory/assignment is a later optional increment. It must never automatically dispatch teams, order supplies, or choose emergency water delivery. Statistical candidate status alone cannot determine operational resource allocation.

If included, tasks have stable ID, investigation ID, title, category, status, assigned responsible role/team (not assumed user identity), due date/time zone, notes, created/updated actor/time, and a change history. Proposed task transitions: `open` → `in_progress`, `blocked`, `completed` or `cancelled`; `in_progress` → `open`, `blocked`, `completed` or `cancelled`; `blocked` → `open`, `in_progress` or `cancelled`; `completed` and `cancelled` are terminal unless an authorised, audited explicit reopen operation is implemented. Every transition is validated server-side. Inventory uses user-entered resource type, known quantity, unit, update timestamp and provenance. Assignment requires a task, resource type, nonnegative quantity and known stock. If requested quantity exceeds known available stock, block the assignment or require correction; do not silently clamp or imply additional stock. Concurrent assignments must not oversubscribe known inventory. Warn when quantity is unknown or stale. Every create/update/assignment records who and when if authenticated identity is available; no actor should be fabricated. Automated optimisation and external dispatch integrations are out of scope.

26. External Dependencies and Integrations

Dependency	Responsibility	Fallback / TBD
Next.js	Web UI/routing/rendering	Verify installed version and supported cache/dynamic-loading APIs
FastAPI / Python	Typed API, validation, deterministic epidemiology and report orchestration	Verify package/runtime versions in repository; unavailable version details are TBD
MongoDB	Metadata, immutable dataset versions, participant records, snapshot metadata	Deployment/topology, transaction availability and backup setup TBD
Groq API	Optional hosted LLM inference	Exact configured model ID/account policy TBD; core features remain functional without it
LangChain / LangGraph	Bounded optional AI graph	Keep graph simple and versioned; package versions TBD
Map provider/library	Optional source-location rendering	Provider and terms TBD; no map required without validated coordinates
PDF/document renderer	Server-side field brief	Select based on existing dependencies, deployment compatibility, licensing, testability and text/table rendering
Hosting/secrets	Frontend/API/database connectivity	Hosting and secret manager TBD; no hard-coded secrets

All third-party integrations require explicit configuration, failure tests and review of privacy/licensing terms relevant to real use. No credentials, supported model ID, API provider URL, package version or hosting provider is asserted by this specification unless later confirmed in repository/deployment records.

27. Testing Strategy

Testing levels: deterministic unit tests for formulas, intervals, ranking and row validation; property/boundary tests for null/zero/overlap/unknown cases; API integration tests for auth, import transaction, version selection, error contracts and cache isolation; frontend component/accessibility tests for forms, state labels and numeric presentation; performance tests for the declared dataset sizes; and end-to-end tests for upload-to-report plus provider failure. Synthetic fixtures are the primary repeatable person-level test data. Published references are aggregate-only benchmarks with explicitly limited use. Real participant data must not be included in test logs or source control.

Use independent hand-checked expected values for synthetic and CDC calculations. Test source-specific tables separately from global totals. Fix seeds and time cutoffs in tests where generated fixtures are unavoidable. Tests of caching must vary dataset version, filter version, method version, and timeline cutoff one at a time. Timeline tests must add a record whose `recorded_at` is one tick after cutoff and prove it changes current results but cannot alter the earlier replay. Security tests must explicitly attempt cross-investigation access, malicious CSV values, formula injection, malformed LLM JSON, unknown source IDs and prompt-injection text. Performance results include measurement configuration; targets are not declared met until measured in the target environment.

28. Acceptance Criteria and Definition of Done

28.1 User-acceptance requirement register

ID	Priority	Observable acceptance behaviour	Verification
ACC-001	MUST	An authorised user creates/ selects an investigation and can identify provenance and current dataset state	E2E investigation workflow
ACC-002	MUST	A valid CSV is previewed, its errors/warnings reviewed, and only an explicitly confirmed accepted set is committed	E2E import test + row reconciliation
ACC-003	MUST	Data-quality counts expose rejected/warned rows and unknown illness/exposure without reclassification	Import/statistics/UI tests
ACC-004	MUST	Source counts, attack rates, RR, CI availability/reasons and status match independent deterministic tests	Stats unit/API/UI tests
ACC-005	MUST	Candidate statuses have explicit reasons and are never phrased as confirmed contamination based solely on association	Ranking/report acceptance review
ACC-006	MUST	Historical replay excludes every record after selected UTC `recorded_at` cutoff and recomputes all outputs	Future-record regression test

ID	Priority	Observable acceptance behaviour	Verification
ACC-007	MUST	Report metric values/status/version/cutoff equal the selected analysis snapshot; report is private	Snapshot equality + access test
ACC-008	MUST	Groq or mapping provider absence does not block deterministic analysis/replay/report; fallback states are usable	Failure-injection E2E
ACC-009	MUST	Core UI meets the approved palette, global styling, responsive behaviour and WCAG 2.2 AA verification plan	CSS/accessibility/responsive tests
ACC-010	MUST	Upload-to-analysis-to-timeline-to-report workflow has automated end-to-end coverage and setup/test documentation	CI report + documented manual review

Implementation is complete only when the following observable criteria pass and are documented:

- A valid CSV can be selected, previewed, validated, reviewed and committed as a version; its schema/hash/provenance/counts are recorded.
- Invalid rows and warnings are shown with actionable locations; no source row is silently changed or dropped from the validation accounting.
- Known exposure + illness records form the expected 2×2 table; unknown exposure and unknown illness are reported distinctly and excluded according to Section 16.
- The synthetic cohort total is 240 with 72 ill/168 not ill, and all three source marginals/RRs/rates match Section 16.5 exactly when the fixture is available or created for testing.
- CI method and zero-cell behaviour match the approved method version and independently computed values; no NaN/Infinity or invented interval is returned.
- Candidate status/ranking is deterministic, explainable and does not imply contamination.
- Timeline cutoff includes records with `recorded_at <= T` only and fully recomputes metrics/statuses; future rows cannot affect historical replay.
- Immutable dataset versions and cache keys prevent stale/cross-investigation result reuse.
- Map has no fabricated coordinates and degrades to a source table when unavailable.
- UI follows the four-token High-Contrast Slate palette, global stylesheet, Heading/Body typography roles, WCAG 2.2 AA target and responsive states.
- Heavy optional views are lazy-loaded with fallbacks; core first-screen content remains available.
- The report’s dataset version, cutoff, counts, metrics, CI and statuses equal the selected snapshot; aggregate reference data are not represented as person-level data.
- The deterministic workflow operates without Groq credentials/provider availability; malformed AI output is rejected and not authoritative.
- Cross-investigation authorization, safe error/logging, upload limits and private response policies have tests.
- Setup, env configuration (without real secrets), tests, deployment and limitations are documented.

29. Deployment and Configuration Requirements

The deployment arrangement is proposed, not repository-verified: Next.js serves the web UI; FastAPI serves the versioned API; MongoDB is a separately configured persistence service; Groq and any mapping provider are outbound integrations. Use HTTPS at public boundaries, server-only secret injection, separate development/test/demo/production settings, health/readiness probes, request timeouts, structured logs and measured resource limits. Network routes should allow only required service-to-database and outbound-provider traffic. No application secrets belong in source control or a browser-visible variable.

Exact hosting services, container/orchestration choice, MongoDB topology, frontend/API domains, CORS origins, authentication flow, backup/restore, TLS termination and deployment commands remain TBD until repository and target environment are available. Do not mandate Docker; use the existing project's deployment conventions if found. Document supported Python/Node runtimes from real lock/config files at implementation time rather than inventing versions. Before real health data is used, a deployment review must confirm authentication, isolation, access logging, backup/restore, retention, HTTPS, incident response and provider privacy terms.

A `.env.example` may list variable *names and descriptions only*, never real credentials. Required variable names should be chosen during implementation from the actual deployment contract. Groq's key must be server-only and must not use `NEXT_PUBLIC_`; database connection strings and session secrets are equally sensitive. Startup must fail clearly or disable optional integration if required configuration is missing, rather than using a fake default credential.

30. Limitations and Risks

1. Observational source-by-source association is vulnerable to recall bias, exposure misclassification, confounding, co-exposure and selection bias; it is not causal proof.
2. A provisional sample-size threshold and Katz log CI are only proposed defaults until reviewed by qualified epidemiology staff. Small/zero cells weaken interval interpretation.
3. User-provided or synthetic data can be incomplete/biased; the interface must expose data-quality and provenance rather than manufacture confidence.
4. Aggregate published benchmarks validate only those aggregate arithmetic examples, not person-level imports, causality or water quality.
5. An LLM may hallucinate or be prompt-injected; bounded payloads, schema/claim validation, human review and non-LLM fallback are mandatory.
6. Real health data create confidentiality and re-identification risks; MVP access/retention/provider policies need approval before deployment with real data.
7. Repository implementation, dependency versions, current schema and existing synthetic fixture were unavailable for inspection in this workspace; all implementation status is therefore unverified.
8. External map/PDF/provider outages must not block deterministic analysis; any external terms, data residency or costs need review.
9. Performance thresholds are proposed targets that require benchmark evidence on the chosen deployment.
10. Operational response resource decisions require verified inventory and authorised human judgment, not automated priority labels alone.

31. Open Decisions and TBD Items

Each recommendation below is proposed, not approved. Core behaviour is specified even where integration choices remain open.

ID	Decision	Why it matters	Recommended default / required approver	Status
DEC-01	Authentication/session method	Controls access to sensitive investigation data and API/exports	Choose a supported server-verified session/ token design after repository/deployment review; security owner approval before real data	TBD
DEC-02	Map provider and terms	Tiles/geocoding terms, privacy, availability and costs differ	MVP source table first; choose provider only if validated source coordinates and a need exist	TBD
DEC-03	CSV limits	Bound memory/CPU and shape expected field imports	Start at 30 MiB, 100k rows, 100 columns, 20 sources; measure target environment then approve	Proposed default; TBD approval
DEC-04	CI method	Affects scientific interpretation and zero-cell behaviour	Katz log 95% CI with no continuity correction as specified; requires epidemiology review	Proposed; approval required
DEC-05	Minimum sample sufficiency rule	Priority labels change based on this threshold	Provisional n≥10 known observations in each group plus CI rule; epidemiology review before operational use	Proposed; approval required
DEC-06	Data/report/audit retention and deletion	Health data, backups, reports and audit logs need a governed lifecycle	Define policy with data owner/privacy reviewer; do not infer statutory period	TBD
DEC-07	Audit detail/access roles	Defines actor identity, export logs and change accountability	Minimal authorised investigator/coordinator permissions for MVP; approve policy	TBD
DEC-08	Groq model/account and data policy	Model support, cost, availability, retention/ privacy depend on exact account	Configure exact supported model ID server-side; verify provider terms before sending real health-derived context	TBD
DEC-09	Performance test environment	Makes targets reproducible	Record deployment instance, runtime, browser, dataset and cache state before benchmark	TBD
DEC-10	Hosting/topology	Determines CORS, networking, secrets, TLS, DB topology	Follow current repository if present; otherwise choose simple separable Next.js/ FastAPI/Mongo deployment	TBD
DEC-11	PDF engine	Rendering, pagination, fonts and deployment support	Prefer existing maintained dependency; test selectable text, pagination and tables	TBD

ID	Decision	Why it matters	Recommended default / required approver	Status
DEC-12	Response Planner scope	Can consume hackathon time and adds data/control complexity	Core workflow first; task checklist optional, inventory/assignment later	Proposed scope; TBD
DEC-13	Source metadata authoring workflow	Determines how columns connect to named source records	Explicit source key/display-name mapping during preview; never infer coordinates	Proposed; implementation approval TBD
DEC-14	Repository baseline/current implementation	Needed to preserve existing code conventions and fixture	Re-run repository review against actual checkout; document discrepancies before implementation	Unverified
DEC-15	Supported runtime/dependency versions	Affects API/cache/rendering behavior	Read package manifests and lockfiles; do not infer versions from this SRS	TBD

32. Traceability Matrix

The detailed requirement tables in Section 12 and the verification scenarios in `AquaTrace_Testing_and_Acceptance.md` form the primary requirement/test map. All requirement IDs below are proposed requirements, not current implementation status. “Proposed” means specified for implementation; “TBD” means final decision or installed-environment proof remains outstanding.

Requirement ID / range	Description	Priority	Module	API/data dependency	Verification	Status
FR-001–FR-006	Investigations, metadata, immutable versions, provenance	MUST	Workspace/Import	Investigations, dataset versions	API + DB integration	Proposed; existing code unverified
FR-010–FR-021	CSV validation, preview, commit and row accounting	MUST	Import	Import validation, participant records	CSV/API/E2E tests	Proposed
FR-030–FR-036	Quality dashboard, deterministic stats, explainable status	MUST	Dashboard/Analysis	Analysis, source results	Unit/API/UI tests	Proposed
FR-040–FR-043	Full timeline replay and optional source map	MUST except map SHOULD	Timeline/Map	Source metadata, cutoff snapshots	Regression/E2E/map fallback	Proposed
FR-044–FR-046	Snapshot-consistent private report	MUST	Reports	Report/analysis metadata	Snapshot equivalence/security	Proposed
FR-050–FR-054	Bounded optional AI summary	MUST to remain optional-safe	AI panel	Aggregate context/model config	Schema/fault/security tests	Proposed
FR-060–FR-062	Optional tasks/inventory/assignments	MAY; controls MUST if enabled	Response Planner	Task/resource/audit records	Validation/concurrency/audit	Optional/TBD scope

Requirement ID / range	Description	Priority	Module	API/data dependency	Verification	Status
FR-070–FR-072	Loading/error states, safe common API errors and idempotent retry behaviour	MUST	Cross-cutting UI/API	API error envelope and request state	UI/API/failure tests	Proposed
DATA-001–DATA-006	CSV, provenance, versioning and entity validation	MUST	Import/data layer	Data dictionary collections	Parser/schema/DB tests	Proposed
STAT-001–STAT-007	2×2 tables, AR, RR, CI, status and timeline replay	MUST	Stats engine	Participant/source records	Independent unit/property tests	Proposed; method approval TBD
AI-001–AI-004	Allowlisted context, output schema, prompt isolation, fallback	MUST when AI enabled	AI graph	Analysis snapshot/config	Unit/integration/fault injection	Proposed
UI-001–UI-005	Four palette tokens, global CSS, typography, accessibility	MUST	Design system	Frontend tokens/components	CSS audit, responsive and WCAG tests	Proposed
PERF-001–PERF-006	Measurable runtime targets	SHOULD	Frontend/API	Fixture/environment	Load/browser tests	Proposed targets; unmeasured
PERF-007–PERF-008	Oversized-input resource safety and mandatory lazy loading	MUST	API/frontend	Upload parser and code split boundaries	Adversarial/resource + bundle tests	Proposed; unverified
SEC-001–SEC-012	Secrets, controlled access, isolation, logs, export controls	MUST	All	Auth/session/audit config	Security tests/review	Proposed; deployed state unverified
NFR-001–NFR-015	Reliability, privacy, availability and cache correctness	MUST	Cross-cutting	Cache policy/version keys	Fault and cache-invalidation tests	Proposed
ACC-001–ACC-010	User-visible acceptance criteria mapped in test plan	MUST for core workflow	End-to-end	API/data/snapshot/report	E2E/manual acceptance	Proposed; not executed

Requirement status convention: `existing` requires inspection evidence, `verified` requires a repeatable check/result, `proposed` is this specification's target, and `TBD` means open. No requirement is marked existing/verified merely because it is written here.

33. References and Supporting Appendices

33.1 External references

1. Centers for Disease Control and Prevention (CDC), *Field Epidemiology Manual*, "Analyze and Interpret Data," table with Sancaktepe Village tap-water exposure aggregate: https://www.cdc.gov/field-epi-manual/php/chapters/analyze-interpret-data.html. Used only for the cited aggregate counts, attack rates, RR and confidence interval. The reference is not a participant-level dataset and is not labelled as cholera.

2. *Cholera outbreak and water resources: Linkages between consumption, infection spread at Mehar village, Sagar district in India*, PubMed Central full text: https://pmc.ncbi.nlm.nih.gov/articles/PMC13177150/. Used as the published aggregate daily-count timeline stated in Section 16.7, not as person-level exposure evidence.
3. World Wide Web Consortium (W3C), *Web Content Accessibility Guidelines (WCAG) 2.2*: https://www.w3.org/TR/WCAG22/. Target conformance level: AA.
4. RFC 4180, *Common Format and MIME Type for Comma-Separated Values (CSV) Files*: https://www.rfc-editor.org/rfc/rfc4180. The import contract is compatible with conventional quoted CSV; the application must still define its strict schema and limits as above.

Framework/API versions and deployment practices are deliberately not cited as fixed version-specific implementation facts because no repository manifests or lockfiles were available to inspect. Verify official documentation for the exact installed versions when implementing.

33.2 Supporting documents

- `AquaTrace_Architecture.md` — component responsibilities, boundaries and data flows.
- `AquaTrace_API_Specification.md` — proposed endpoints, request/response structures and errors.
- `AquaTrace_Data_Dictionary.md` — entity fields, validation, collection links and index guidance.
- `AquaTrace_Testing_and_Acceptance.md` — test catalogue, edge cases, fixtures and requirement traceability.

These support the SRS; where a conflict is found, preserve the product-safety/statistical constraints in the SRS and update the conflicting supporting document before implementation. None is evidence that application code currently exists or passes tests.